Recommended 6-Week Roadmap
10–15 hours/week
Goal: Build production-style AI integrations with Django.

Week 1: Python Engineering Fundamentals

Topics

Object-Oriented Programming

- Classes
- Objects
- Methods
- Constructors
- Encapsulation
- Inheritance
- Composition

Advanced Data Structures

- Dictionaries
- Sets
- Tuples
- Nested structures
- List comprehensions
- Dictionary comprehensions

Clean Code

- Functions
- Type hints
- Docstrings
- PEP 8

Error Handling

- try/except
- custom exceptions
- logging basics

Project

Contact Management System

Build:

- Create contact
- Update contact
- Delete contact
- Search contact
- Save/load data

Focus:

Code organization
Classes
Error handling

Week 2: Files, APIs, JSON, and External Libraries
Topics
Files

- Text
- CSV
- JSON

APIs

- HTTP requests
- GET
- POST
- Status codes

External Libraries

- requests
- python-dotenv

Environment Variables

- API keys
- Secrets management

Project
Customer Data Processor
Reads:

Customer file

Calls:

Public API

Produces:

Business report

Focus:

JSON
API communication
Data transformation

Week 3: Databases and Application Architecture

Many AI beginners skip this and regret it later.

Topics

SQL Review

- Joins
- Indexes
- Relationships

Django ORM

- Models
- Queries
- Relationships

Application Architecture

Create layers:

views
services
repositories
models

Logging

- Application logs
- Error logs

Project

Customer Support Ticket System

Features:

- Create tickets
- Assign priorities
- Store history

Focus:

Database design
Architecture
Maintainability

Week 4: AI Fundamentals and LLM Integration

Now we finally introduce AI.

Topics

LLM Fundamentals

- Tokens
- Context windows
- Temperature
- System prompts

Prompt Engineering

- Summarization
- Classification
- Extraction
- Rewriting

Structured Outputs

Learn why businesses prefer:

```
{
  "category": "",
  "priority": "",
  "summary": ""
}
```

instead of free text.

Project

AI Ticket Analyzer

Input:

Customer complaint

Output:

Priority
Category
Summary

Focus:

Reliable AI responses
Prompt design

Week 5: AI Business Workflows

This is where AI starts creating business value.

Topics

AI Workflow Design

Patterns:

Input
→ AI
→ Validation
→ Database
→ User

Multi-Step Processing

Examples:

Document
→ Extract data
→ Classify
→ Store

Cost Awareness

Learn:

- Token usage
- Pricing
- Caching

Security

- Prompt injection basics
- Input validation

Project

AI Lead Qualification Tool

Input:

Lead inquiry

AI determines:

Interest level
Urgency
Product fit
Follow-up recommendation

Week 6: Production AI Applications

Most tutorials stop before this stage.

Topics

Django + AI Integration

Build proper service layers.

Monitoring

Track:

- Requests
- Errors
- AI usage

Reliability

- Retries
- Timeouts
- Fallbacks

Deployment Concepts

Learn:

- Docker basics
- Environment configs
- Production settings

Final Project

AI Business Assistant

Choose one:

- Sales Assistant
- Customer Support Assistant

- Meeting Notes Assistant
- Contract Summarizer
- Knowledge Base Search Tool

Features:

User Management

- Login
- Dashboard

AI Features

- Submit content
- Receive analysis
- Store results

Business Features

- Search history
- Reports
- Usage tracking

What I Would Add If You Continue Beyond 6 Weeks

After this foundation, the next progression would be:

Weeks 7–10

- Embeddings
- Vector databases
- RAG (Retrieval-Augmented Generation)
- Semantic search
- AI agents
- Tool calling
- Function calling
- Multi-agent workflows

These topics are where modern business AI applications are heading, but they make much more sense after you've built several AI-powered applications first.

For your stated goal, I'd structure our sessions as a 6-week curriculum and treat Week 1 as a software-engineering foundation, Weeks 2–3 as application-building skills, Weeks 4–6 as AI integration and production practices. That sequence mirrors how AI products are actually built in businesses today.